



Scope of Work

KHMS & KReS

MS Department
Krishak Bharati Cooperative Limited, Surat



Contents

1.0 Overview	3
1.1 Introduction	3
KRIBHCO Hospital & KHMS	3
KReS	3
1.2 Purpose	4
1.3 Pre-Bid Meeting	4
2.0 Engagement Information	4
2.1 KHMS – Business & Technical Scope	4
Patient Module	4
Nurse Module	5
Laboratory Module	5
Administration Module	5
Medical Store	6
Details of information to be captured	6
Queries	14
Reports	14
Security / Authorizations	15
Download/Synch. Employee data from SAP ERP system	15
General	16
2.2 KReS – Business & Technical Scope	16
List of Modules & Functionalities	17
Download/Synch. Employee data from SAP ERP system	21
Reports	21
Push Notifications	22
Offline Access	22
Native/Hybrid App	22
Device/OS Compatibility	22
Additional Requirement	22
2.3 Statement of Work	22
2.3.1 Design	22
2.3.2 Deliverables	23
2.3.3 Implementation Methodology	23
2.3.4 Deployment Landscape	23
2.3.5 Exclusions	23

2.3.6 Payment Terms	23
2.3.7 Security	23
2.3.8 Post Go-Live Support.....	24
3.0 Proponent Response.....	24
3.1 Introduction	25
3.2 Mandatory Conditions	25
3.2.1 Compliance with KRIBHCO's Terms and Conditions	25
3.2.2 Availability for Immediate Start	25
3.2.3 Commitment to Go-Live Date	25
3.3 Expertise & References	25
3.3.1 Demonstrated Corporate Expertise	26
3.3.2 Corporate References	26
3.4 Proposed Services	26
3.4.1 Client Approach.....	26
3.4.2 Engagement Approach.....	26
3.4.3 Phases, Sub-Phases, Activities and Timing.....	26
3.4.4 KRIBHCO's Roles and Responsibilities.....	26
3.4.5 Proposed Resources.....	27
3.4.6 Proposed Resource Bios.....	27
3.5 Financials.....	27
3.5.1 Pricing.....	27
3.5.2 Exclusions.....	28
3.5.3 Conditions and Assumptions	28
3.5.4 Warranties	28
Annexure.....	29
Home Page	29
Sign In/ Sign Up	30
Sign Up	30
Sign In.....	30
Telephone Directory	31
Circulars	32
PCAMC Complaints	33
Estate Complaints	34
Estate Admin	35
PC Admin.....	36

1.0 Overview

1.1 Introduction

Krishak Bharati Cooperative Limited (KRIBHCO) is a premier National level Cooperative Society of India engaged in fertilizer production and distribution and is registered under Multi State Cooperative Societies (MSCS) Act, 2002. MS Department in KRIBHCO, Surat supports all the IT application, network, storage and infrastructure needs for its employees deployed in Hazira manufacturing unit.

KRIBHCO Hospital & KHMS

KRIBHCO operates a hospital situated at its township inside the KRIBHCO Hazira Fertilizer complex. The Hospital provides basic medical facility to township residents. The hospital is equipped with full time resident doctors, visiting specialist doctors and Nursing staff working in all 3 shifts. There are OPD & IPD (Indoor) facility, medical stores and a Pathology lab operating inside the hospital for the service of patients.

KRIBHCO Hospital Management System (KHMS) is a system developed in the year 2005 using Powerbuilder at front end and Oracle9i as the database. This application is used by the resident & visiting doctors, Nurses and staff to manage all OPD & IPD patients, the medicine inventory, Pathology records, and general administration. However, the present system has reached its end-of-life and needs replacement primarily due to following factors:

- The present system is old and no support is available from vendor for the front-end software Power builder. Upgradation in Oracle may not be supported by the present old version of Power builder.
- The present system is not web enabled.
- Earlier there was only one medical store in the hospital. Now a new medical store also come up. In future the vendor for medical store may change and there needs to be an easy method to set up a new medical store.
- Integration with KRIBHCO's SAP ERP system for accessing employee related details is proposed in the new system.

KReS

KReS (Kribhco e-Seva) application aims to consolidate all non-strategic (non-ERP) business processes such as HR, IT and other general services sought by employees into one web portal and a (iOS & Android) mobile application. At present, we have identified host of services to be automated as described in section 2.2 of this document. The solution must be scalable to be extended to other services as and when such need arises.

1.2 Purpose

The purpose of this RFP document is to invite prospective IT solution providers to provide their best quotation for applications mentioned in schedule below:

SN.	Description	Unit	Quantity
1	Design, Development and Deployment of web-based KHMS (KRIBHCO Hospital Management System) System as per business & technical scope mentioned in Section 2 inclusive of 3 months post go-live support.	Nos.	1
2	Design, Development and Deployment of web-based KReS (KRIBHCO e-Seva) System as per System as per business & technical scope in Section 2 inclusive of 3 months post go-live support.	Nos.	1
3	Design, Development and Deployment of Android & iOS based Mobile Application for KReS KRIBHCO e-Seva) System as per System as per business & technical scope mentioned in Section 2 inclusive of 3 months post go-live support.	Nos.	1

1.3 Pre-Bid Meeting

KRIBHCO shall invite prospective bidders for a **pre-bid meeting** so that they familiarize themselves with existing systems, KHMS & KReS, currently deployed in KRIBHCO and further understand the requirement for the proposed system.

2.0 Engagement Information

2.1 KHMS – Business & Technical Scope

KRIBHCO Hospital Management System (KHMS) shall have the following modules:

1. Patient Module
2. Nurse Module
3. Laboratory
4. Administration
5. Medical Store

Patient Module

The Patient module shall be operated by the Doctors / Office Staff for the following purposes:

- To Maintain various masters such as
 - Patient Master
 - Doctor Master
 - Sign & symptom Master
 - Disease Master
 - Dose Master
- To Create OPD Cases & Prescribe various Treatments
- To create Reference Letter for outside treatment (to empanelled hospitals & doctors)
- To issue Sick Fit Certificate
- To record & issue Death Certificate
- To retrieve various reports from the system

The proposed system shall have a provision for doctors to retrieve patient's treatment history, Today's cases at the time of case registration.

Nurse Module

The Nurse module shall be operated by the Nursing Staff for the following purposes:

- To register OPD Cases & issue medicines in non-OPD hours
- To register Indoor Patients & issue medicines as well as referring Tests to Lab
- To retrieve report for analysis & issue to various agencies

The proposed system shall have a provision for Nursing staff to retrieve patient's treatment history at the time of case registration.

Laboratory Module

Laboratory Module shall be used by Pathology Section staff to record test results prescribed by doctors as well as Nursing Staff. Test Result report is also issued from the Laboratory module to the patients.

Administration Module

The Administration module shall be used by the Hospital Administration Staff for the following Purposes:

- To Maintain various masters as below:
 - Specialisation Master
 - Surgical Treatment Master
 - Patient Relation Master
 - Patient Grade Master
 - Category Master
 - Hospital Master
 - Ward Master

- Test Master
- Section Master
- Employee - Section Master Mapping
- Medicine Master
- Sectional Medicine Master
- Charge Master
- Parameter Master
- To assign various screen authorisations to KHMS users
- To set up user account for new Doctors/users in the system

Medical Store

This module shall be used by medical store staff from where the medicines are issued to patients as per the prescription of Doctors. These stores are set up inside the hospital and operated by Private Parties. At present there are two medical stores in the hospital. There can be more medical stores in future.

The module shall be used for the following purposes:

- To upload medicines / stock into KHMS system as and when required, normally once in a day.
- To modify sectional item master details
 - To update stock individually
 - To enter new batch stocks for a particular medicine
 - To update the expiry date for a batch

The proposed system shall provide an easy way to configure and set up new medical stores and when the requirement arises.

A proven and easy mechanism may be set up to upload medicine details of the private parties into the proposed system and the uploaded stock is available to doctors/nurses for prescription and reference.

Only selected tables shall be made available to the Private vendor who run the Stores for uploading medicine stocks.

Individual stores shall be allowed to see & maintain their own data only.

Details of information to be captured

Master Data

1. Patient Master
 - Patient Code
 - Name
 - Sex
 - Category (Employee, Dependent, Non-Dependent, Emergency Patient etc.)
 - Address

- Employee Pers No
 - Eligibility Grade
 - Relation with the employee
 - Date of birth
 - Blood Group
 - Contact Details
 - High Risk Patient
 - Status (Active/Terminated/Superannuated...etc)
 - Status Date
 - Date of Retirement / Treatment Permit Date
 - Allergy Details (Disease Code from Disease Master) - Multiple
 - High Risk Details (Disease Code from Disease Master) - Multiple
2. Doctor Master
- Doctor ID
 - Doctor Type (KRIBHCO/ Visiting/Outside Reference)
 - Employee P No in case of KRIBHCO Employee)
 - Name
 - Sex
 - Address
 - Specialization
 - Hospital Name & Address
 - Contact details
 - Tax Exempted Flag
3. Sign & symptom Master
- Sign symptom ID
 - Sign symptom Name
 - Specialization (Multiple)
4. Disease Master
- Disease ID
 - Disease Name
 - High Risk Flag
 - Specialization (Multiple)
5. Dose Master
- Dose ID
 - Dose Name
 - Dose Type (Drugs / Injection etc.)
6. Specialization Master
- Specialisation ID

- Specialisation Name
7. Surgical Treatment Master
 - Surgical Code
 - Surgical Name
 - Type (Minor / Major etc.)
 8. Patient Relation Master
 - Relation ID
 - Relation Description
 9. Patient Grade Master
 - Grade Code
 - Description
 - Entitlement
 10. Category Master
 - Main Group
 - Description
 - Category ID
 - Category Name
 - Category Type
 - Chargeable (Y/N)
 - Outside reference Flag (Y/N)
 11. Hospital Master
 - Hospital Code
 - Hospital Name
 - Address
 - Contact Details
 - Tax Flag
 12. Section Master
 - Section Code
 - Section Name
 13. Ward Master
 - Ward Number
 - Ward Description
 14. Main Test Master
 - Test Code

- Test Name
 - Section Code
15. Sub Test Master
- Test Code
 - Test Name
 - SI Num
 - Main Test Code
16. Medicine / Item Master
- Medicine Code
 - Medicine Name
 - Unit of Measurement
 - Medicine Type (Injection / Medicine / General etc.)
 - On hand Qty
 - Batch (Multiple)
 - Expiry Date (Multiple)
 - Stock Qty (Multiple)
17. Sectional Medicine Master
- Medicine Code
 - Section Code
 - Batch No
 - Expiry Date
 - Stock Qty
 - MRP
 - Discount%
18. Charge Master
- Date Applicable
 - Section (Indoor/OPD/Pathology)
 - Category of Patient
 - Test ID
 - Description
 - Charge
19. UOM Master
- UOM Code
 - Description

Transaction Data

Case Registration (By Doctors / Nurse Room)

- A single screen may be provided to capture all details related to the case.
- Current stock of medicines in all stores may be made available to the doctor for prescription.
- Data selection popup screens may be displayed automatically to reduce data entry work of the user.
- Patient's history of pathological lab results to be displayed as a trend analysis option to doctors.
- Proper search/sort/find facility for data on pop-up screens may be provided.
- Provision to see patient's treatment history may be made available on case registration screen. For this historic data may need to be ported to the new system.
- Provision to see Today's cases may be provided on screen.
- Prompt may be given if the patient is allergic or high risk.
- Patient's basic details may be displayed on entering patient code.
- Patient photograph may be shown on screen on entering Patient Code.
- Patient current details such as category, grade, Employee P No, Registered by Doctor/Nurse, case registered by user id etc. may be stored internally along with the case details.
- Patient's age may be calculated and displayed on screen.
- Once the case is saved, prescriptions may be printed automatically with a confirmation from the user.

1. Case Header – For registration of patient cases

- Case Number (Auto Generated)
- Date & Time
- Doctor ID
- Patient Code
- Special Comments
- Emergency Flag
- Indoor Flag
- Indoor No
- Systolic
- Diastolic
- Pulse

2. Case Sign Details

- Sign ID
- Findings
- Remarks

3. Case Symptom Details

- Symptom ID
- Findings
- Remarks

4. Case Provisional Diagnosis

- Disease ID
- Remark

5. Case Drug Issue details

- Medicine Code
- Dose ID
- Qty
- Remark

Current Stock Qty, batch, price etc. may be stored internally against each issue.

6. Case Injection Details

- Injection Code
- Dose ID
- Qty
- Remark

Current Stock Qty, batch, price etc. may be stored internally against each issue.

7. Case Immunization Details

- Vaccine Code
- Qty
- Given Date

Current Stock Qty, batch, price etc. may be stored internally against each issue.

8. Case Pathology Test Details

- Main Test Code
- Remark
- On selecting Main Test, Sub test details shall automatically be populated.
- Test results shall be entered by pathology section by retrieving the tests prescribed by Doctor/Nurse.

Details selected from Test Master. Sub Test if any also need to be captured.

9. Case Surgical Details

- Surgical Code
- Remark

Surgical Treatment Taken flag shall be updated by Nursing Staff

10. Outside Reference

- Reference Doctor/Hospital
- Emergency Case
- Referred for (OPD/IPD/Xray/Investigation/Consulting/Others)
- Special Comments

On selecting Reference Doctor/Hospital, details of hospital may be displayed on screen.

Provision to print reference letter from Case registration screen may be provided.

Indoor Registration

A case shall be registered either by Doctor or Nursing staff prior to Indoor admission.

1. IPD Header Table (For Indoor Admission)

- IPD Number
- IPD Date
- Case Number
- Admission Date
- Ward Number
- Bed Number
- Remark
- Discharge Date
- Discharge Remark
- Discharge Reason

On selecting case number, case details including diagnosis may be displayed on screen.

2. Treatment Details

- Drug Date
- Medicine Code
- Dose ID
- Qty
- Rate
- Remark

Drug batch wise issue details may be displayed on screen.

3. Pathology Test Details

- Test ID
- Test Date
- Result

4. Observation Details

- Date
- Observation
- Remark

5. Visit Details

- Date
- Doctor ID

- Remark
6. IPD Other Procedure Details
 - Date & Time
 - Procedure Code
 - Procedure
 7. IPD Surgical Details
 - Test Date
 - Surgical Code
 - Result

Surgical Treatment Given Details

- Retrieved by Case Number
- Details of case to be displayed
- Details to capture
 - Treatment Given Flag
 - Treatment Given Date (Populate system date)
 - Remarks

Pathology Test Results - OPD

- Test retrieved by Case No
- Case details to be displayed
- Test Details are populated & Results entered by Pathology
- A pathological trend of results from history of past pathological report to be displayed
- Details to be captured
 - Test Date
 - Result
 - Abnormal Flag
 - Remarks

Pathology Test Results - IPD

- Test retrieved by IPD No
- Case details to be displayed
- Test Details are populated & Results entered by Pathology
- A pathological trend of results from history of past pathological report to be displayed
- Details to be captured
 - Test Date
 - Tested Flag
 - Result
 - Abnormal Flag
 - Remarks

Sick Fit Certificate

- Date

- Doctor ID
- Patient Code
- From Date
- To Date
- Resume Date
- Remark

Death Certificate

- Date
- Registration Number
- Registration Date
- Doctor ID
- Patient Name
- Age
- Requestor Name
- Requested Date
- Reason
- Death Date
- Death Place
- Sex

Queries

The following reports shall be available from the system:

1. Master Data Query with Export to Excel
 - Patient List
 - Doctor List
 - Specialisation Details
 - Disease List
 - Dose List
 - Sign / Symptom List
 - Charge Details
 - Medicine Master details
 - Surgical Treatment Master details
 - Xray Master details

Reports

1. Prescription (Medical Store wise for Case No)
2. Reference Letter
3. Medical Expense Report (Patient wise / Category wise) with export to Excel

4. Lab Test Report
5. Drug Issue details by Date Range, Medical Store wise
6. Disease wise Patient List
7. Doctor wise Patient seen
8. Outside reference for a period

Maximum 15 Nos reports

Security / Authorizations

Administration section will be responsible for creating new users and providing necessary authorizations. Only limited persons shall be given authorization to use the Security module.

- User Groups can be created with access to selected modules / screens in the system. These user groups can be assigned to individual users.
- Section wise active users can be maintained in the system. At a time, a user can be active in one section only (For example, Nursing staff will be active in the Nurse Section Only). User shall get the permission to use the screens in the active section only.

Following permissions can be assigned to these groups:

- All
 - Read Only
 - Modify Only
 - Insert Only
 - Insert & Modify
- Individual screen permission can also be assigned to any user. Following screen permissions can be assigned:
 - All
 - Read Only
 - Modify Only
 - Insert Only
 - Insert & Modify

Download/Synch. Employee data from SAP ERP system

A mechanism shall be developed to integrate HR Employee data in to the application. A program shall be developed in SAP (KRIBHCO's scope) to import the employee master in desired format (csv, excel, text). An interface in KHMS app will then facilitate upload of employee master periodically from above file. The imported data shall be linked with user profiles on the application.

General

- Uniform template may be used to create all user interfaces.
- Retrieval window may be provided on all screens.
- Search, Sort, find facility may be provided on child data screens and pop-up screens.
- System generated numbers may be used as Primary key fields wherever required as suggested at system study.
- Provision to notify patients on generation and publish of their pathological report.
- Medicines may be issued in FIFO basis.
- Expired medicines shall not be shown or issued.
- Expired medicines shall not be allowed to upload into system by Vendor. List of checks to be done at the time of upload of medicines shall be provided.
- Export data to Excel facility may be provided in Reports / Query wherever needed.
- Registered cases are modifiable by the doctors for the current date only. In case any modification is required subsequently, the doctor need to contact the administrator.
- Registered cases can be modified by the Nurse section for 1 hour only. In case any modification is required subsequently, the doctor need to contact the administrator.
- Pathology tests (OPD) are modifiable by the staff for 2 days only. In case of IPD patients, the days shall be calculated from the proposed test date for individual Test Codes.

2.2 KReS – Business & Technical Scope

Following Modules and Functionalities have been identified to be included in the application

Module	Mobile App	Website	Description
User Authentication	✓	✓	Sign In/ Sign Up, Password Reset
User Profile	✓	✓	Manage User Personal Information & Password Change
User Management	✗	✓	User, Roles and Authorization Management
IT Support	✓	✓	Create/View IT Incidents
IT Support Management	✗	✓	IT Incident Management
Estate Complaint	✓	✓	Create/View Township Maintenance Incidents
Estate Management	✗	✓	Township Maintenance Incident Management
Guest House Booking	✓	✓	Create/View Guest House Booking
Guest House Management	✗	✓	Guest Requisition Management (Approval/Rejection)

Vehicle Requisition			Create/View Vehicle Requisition
Vehicle Administration			Vehicle Requisition Management (Approval/Rejection)
Notices & Circular Publication			Display Notices/Circular pdf/image/blog
Notices & Circular Admin			Create/Upload Notices/Circular
Telephone Directory			Display Telephone Directory

List of Modules & Functionalities

User Authentication

A mechanism must be adopted to authenticate users on KReS website and Mobile Application using their Personal Number as Login ID and some form of Password.

User Profile

This module shall be used to display the User Master data such as:

- Personal Number
- Name
- Designation
- Telephone Number (shall be used in Telephone directory module)
- Email Address
- Quarter Number (shall be used in Estate management module)

User Management

This module shall allow administrators to download and synch the user master data from SAP HCM Module. Further it will allow administrators to manage (create, delete & modify) user data and assign them relevant authorization (user, super admin, IT Support Admin, HR admin, Estate admin, etc.) This being an admin module is required only on the KReS website.

IT Support

This module shall be used by users to solicit IT support for their allotted Hardware. Following fields shall be captured for registering an IT support request:

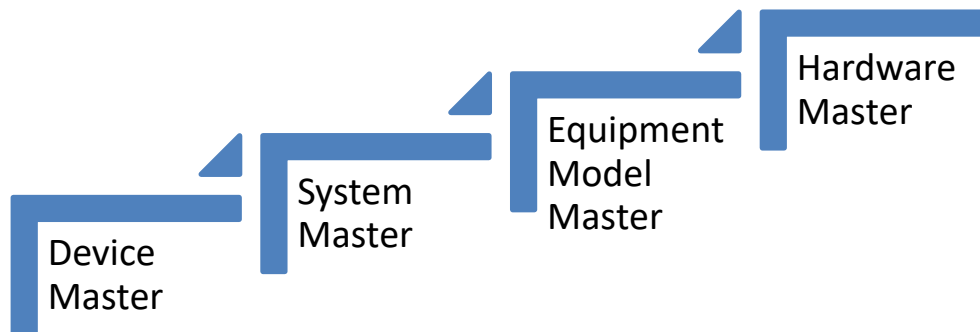
- Machine Code
- Complaint Number (auto generated)
- Problem Description
- Reported Person
- Reported Date & Time
- Complaint Status

IT Support Management

This module shall allow IT administrators to manage IT assets, IT Support requests and IT support AMC. The IT support at KRIBHCO is outsourced via an AMC contract and most of its IT asset are covered under the AMC. This module should facilitate master data management such as Hardware master and transactions such as IT asset allocation, IT support ticket resolution etc.

Master Data

1. Hardware Master must be maintained to capture Hardware details. The current system has a hierarchical hardware master structure as represented below:



Some of the fields captured in Hardware Master include

- Machine Code
 - Department Code
 - Device Code
 - System Name
 - Model
 - Make
 - Location
 - Stand Alone (Y/N)
 - Vendor Code
 - Inst. Date
 - Exp. Date
 - AMC From
 - AMC To
 - Kribhco Hardware (Y/N)
 - Days under Maintenance
 - Status
 - Machine SL No.
 - Current Rate
2. Vendor Master – to capture details of PC AMC contract vendor.
 3. AMC Rate Master – to maintain the AMC rate for each Device/system.

Transactions

1. IT Support Request - Following fields should be captured to respond to IT support request:
 - Complaint No.
 - Complaint Type
 - Machine Code
 - Reported Date & Time
 - Contact Person (Complainant)
 - Call Report No.
 - Category
 - Action Taken
 - Engineer Name
 - Expected Completion Date
2. IT Hardware allocation

Each IT support request resolved under AMC will involve cost for each transaction that shall be calculated using AMC Rate master. This being an admin module is required only on the KReS website.

Estate Complaint

This module shall be used by users to lodge Civil/Electrical complaints for their Quarters or Public Buildings. Following fields shall be captured for lodging Estate Complaints:

- Complaint Type (Civil/Electrical)
- Work Type
- Building Type (Quarter/Public Building)
- Quarter Number
- Work Requested
- Complaint Date Time
- Reported Person
- Complaint Status

Estate Management

This module shall allow Estate administrators to manage Estate Complaints. Following fields should be captured to respond to:

- Complaint Type (Civil/Electrical)
- Work Type
- Building Type (Quarter/Public Building)
- Quarter Number
- Work Requested
- Complaint Date Time
- Reported Person
- Complaint Status
- Complaint Attended By
- Complaint Close Date Time

Further a quarter master must be maintained with relevant fields to fetch quarter details. This being an admin module is required only on the KReS website.

Guest House Booking

This module shall be used by users to requisition a guest house facility at KRIBHCO Township's Hazira Fertilizer Complex. Following Details must be captured to requisition a guest house:

- Document Date
- Guest House Type (I, II, III)
- Description of Guest
 1. Name & Address of Guest
 2. Date-Time of Arrival & Departure
- Purpose of Requisition (Personal, Official)
- Requirement of Rooms (Nos)/Dormitory/TV Room/Lounge
- Instructions for Guest's House Hospitality
 1. Payment of lodging charges (Personal/Official)
 2. Payment of Boarding charges (Personal/Official)
 3. Type of Food (Veg, Non-Veg, Ordinary, Special, VIP)
- Special Instruction
- Applicant's Detail
 1. Name
 2. Designation
 3. Personal No.
 4. Department
 5. Intercom No.
- Approving Authority's Details
 1. Name
 2. Designation

Guest House Management

This module shall be used by HR Admin to view and respond to Guest house requisitions with Approve/Reject status. This being an admin module is required only on the KReS website.

Vehicle Requisition

This module shall be used by users to requisition a vehicle from HR Department. Following details must be captured:

- Document Date
- Requester's Details
 1. Employee's Name
 2. Designation
- Personal Number
- Vehicle Request Description
 1. Date
 2. Time

- Pick-up location
- Purpose of Request
 1. (Personal/Official)
 2. Reason for Travel
- Destination
- Approximate Duration of Travel
- Applicant's Detail
 1. Name
 2. Designation
 3. Phone No.
- Approving Authority's Details
 1. Name
 2. Designation

Vehicle Administration

This module shall be used by HR Admin to view and respond to Vehicle requisitions with Approve/Reject status. This being an admin module is required only on the KReS website.

Notices & Circular

This module shall be used to display Notices & Circulars released by management in a user-friendly article-based interface.

Notices & Circular Administration

This module shall be used by HR administrator to manage content of Notices and Circulars for publication to users.

Telephone Directory

This module shall be used by users to view Contact details of employees working in KRIBHCO Hazira plant. Following fields shall be displayed in the telephone directory:

- Employee personal number/Employee ID
- Employee name
- Employee mobile number
- Office phone number
- QTR number
- Residence phone number

Download/Synch. Employee data from SAP ERP system

A mechanism shall be developed to integrate HR Employee data in to the application. A program shall be developed in SAP (KRIBHCO's scope) to import the employee master in desired format (csv, excel, text). An interface in KReS app will then facilitate upload of employee master periodically from above file. The imported data shall be linked with user profiles on the application.

Reports

Reports must be available for analysis of data. Some of the reports needed in KReS include:

- IT Support Module
 - Hardware List (Based on criteria such as Device, System, Equipment Model, Department)

- Complaint Register (Based on Hardware code, Department)
- AMC Billing (Based on Vendor, Date, Device)
- Estate Management Module
 - Estate Complaint List (Based on Complaint No., Date, Quarter Number, Status, Category)

Push Notifications

Push Notifications shall be built in to the mobile app for notifying users primarily about fulfillment of their IT support, Estate, Guest House, Vehicle requisitions and addition of new Circulars/Notices, etc.

Offline Access

The app must be capable of offline access to some critical applications such as Employee Directory and Circular/Notices. The app should have a built-in storage to store latest records from Employee Directory and Circular/Notices modules whenever the device is in synch with the internet. The latest synched records shall be available for offline access.

Native/Hybrid App

The proponent can propose either a Native or a Hybrid Cross platform approach to mobile app development.

Device/OS Compatibility

The mobile application must support the latest version, as on date of go-live, of Android and iOS operating systems. Further the app must also support all devices running on iOS and Android OS versions released since 01 January 2018.

Additional Requirement

The proponent shall propose a solution which in addition to the above must also offer continuous life cycle management, API Management, built-in security and security mediation, versioning and easy deployment between environments (Development, Staging and Production).

2.3 Statement of Work

2.3.1 Design

- The business requirement contained in Section 2.1 & 2.2 for KHMS & KReS outlines a broad scope of work. The actual number of fields, tables, reports, screens & processes may vary based on actual requirement gathering carried out by successful proponent after analysing the present systems and interacting with field officers and staff.
- Requirement gathering workshop shall be carried out by interacting with KRIBHCO hospital staff as well as other core users and studying the existing system. Feedback and suggestions from the staff shall be sought for any new requirement in the proposed system.
- A high-level design document shall be created based on Requirement study carried out by successful proponent. The low-level design will be developed based on the finalized high level design document.
- A business blueprint document comprising high-level and low-level design shall be developed to freeze the scope of work and signed off by KRIBHCO & the proponent.

2.3.2 Deliverables

During the design phase:

- Software High-level design document
- Software Low-level design document
- Business Blueprint

During & After Execution:

- Technical build including source-code of front-end and back-end platforms
- Unit testing
- Support for integration, end to end and UAT testing
- Complete documentation of the project including, but not limited to installation, configuration, user manual, system manual, etc.
- Support for deployment and hosting
- Warranty period support
- Providing Training to our end users, Business Power users, IT Administrators and IT Developers.

The successful proponent shall hand over 100% source-code of software and graphics to KRIBHCO for the purpose of copyright and intellectual ownership. The successful proponent shall not have the right to use the Intellectual Property and the transferred Software.

2.3.3 Implementation Methodology

- The work shall be carried out in hybrid mode both at KRIBHCO's and vendor's premise by accommodating necessary suggestions from MS department.
- Testing of the system shall be done with the help of core users comprising Hospital staff, HR & IT department officials.
- The complete system after development shall be handed over to MS department.
- Existing data in present KHMS & KReS system to be migrated to the new system.

2.3.4 Deployment Landscape

The successful proponent shall submit deployment plan during the Business Blueprint stage. The plan of deployment must consider options of full stack deployment either on premise or on cloud. The final decision shall be taken by the KRIBHCO management based on operational and economic concerns.

2.3.5 Exclusions

Hardware & Software such as servers, Database, OS licenses required for deployment of application shall be in KRIBHCO's scope however, technical support required for successful deployment shall be in vendor's scope.

2.3.6 Payment Terms

The payment schedule shall be divided in milestones broadly consisting of Design phase, Implementation phase, Go-Live and Post Go-Live support.

2.3.7 Security

KRIBHCO holds security with regards to unauthorized access of data of utmost importance and we expect the proponent to design the app placing application security as its prime concern. The application architecture must focus on following security aspects:

- Application should employ Data encryption for communication over network.
- Manage user's application sessions and session time out.

- Limit the total number of simultaneous sessions from the same user.
- The application must be build based on Secure Coding Standards.
- Perform content validation on the web application to ensure the web application is not vulnerable to phishing attacks.
- The Application must have facility for Two Factor authentication using hardware or software token.
- SSL certificate must be deployed for the application.

2.3.8 Post Go-Live Support

The vendor shall provide service for incidents arising in the delivered KHMS & KReS application which:

- Leads to abnormal functioning of the system and in deviation to agreed business process, such as a bug which prevents normal operation of system.
- Leads to an error which stalls the normal operation of any transaction, such as system crash or any exception to normal flow of transaction.

In addition, the vendor shall provide services for:

- Update of software application versions such as database, middleware, web application container, IDE or any other software used for development, testing and deployment of KHMS & KReS application.
- Creation of new Reports for KHMS based on users' requirement. The total number of reports in KHMS not to exceed 20 Nos including those developed before Go-Live.

3.0 Proponent Response

The following section describes the **mandatory** content of the response to this RFP. Proponents are advised to respond in the order outlined in the table of contents below.

Proposal Section	Proposal Table of Contents	RFP Section Reference
1.	Introduction	3.1
2.	Mandatory Conditions	3.2
	a. Compliance with KRIBHCO's Terms & Conditions	3.2.1
	b. Availability for Immediate Start	3.2.2
	c. Commitment to Go-Live Date	3.2.3
3.	Expertise & References	3.3
	a. Demonstrated Corporate Expertise	3.3.1
	b. Corporate References	3.3.2

4.	Proposed Services	3.4
	a. Client Approach	3.4.1
	b. Engagement Approach	3.4.2
	c. Phases, Sub-phases, Activities & Timing	3.4.3
	d. KRIBHCO's Roles & Responsibilities	3.4.4
	e. Proposed Resource(s)	3.4.5
	f. Proposed Resource Bios	3.4.6
5.	Financials	3.5
	a. Pricing	3.5.1
	b. Exclusions	3.5.2
	c. Conditions and Assumptions	3.5.3
	d. Warranties	3.5.4

3.1 Introduction

This section is provided for the purposes of an introductory letter from the Proponent.

3.2 Mandatory Conditions

3.2.1 Compliance with KRIBHCO's Terms and Conditions

Proponents must counter sign and submit their acceptance of KRIBHCO's terms and condition attached along with the RFP.

3.2.2 Availability for Immediate Start

Proponent must explicitly commit to availability to start work immediately upon receiving the contract award.

3.2.3 Commitment to Go-Live Date

Proponent must explicitly commit to carry out Go-Live of KHMS & KReS within the specified time frame. Proponents should advise the KRIBHCO of approaches they will take to meet this commitment.

3.3 Expertise & References

3.3.1 Demonstrated Corporate Expertise

Proponents are required to provide a concise outline of their corporate expertise. The outline should cover the following:

- Corporate value proposition
- Number of years in business, size and geographical spread of corporation, market position.
- Number of employees, outline of client base.
- 2 to 3 case studies demonstrating recent and relevant corporate expertise (no more than 2 pages per case study). Provide at least one example with similar project scope.

Industry designations and/or certifications possessed that are relevant to the initiative may also be noted.

3.3.2 Corporate References

Proponents are required to provide references for recent successful engagements of similar size and scope, and that represent the Proponent's current corporate capability. To verify these references, Proponents must provide a list of contacts and site references. Contacts provided in the proposal should be notified that they may be contacted by the KRIBHCO as part of the evaluation process. References from relevant proponents from fertilizer/chemicals vertical shall be considered more desirable.

3.4 Proposed Services

3.4.1 Client Approach

Proponents are asked to describe the approach they take to working with clients, including but not limited to: understanding client needs; working with client stakeholders; governance and oversight; issue and risk management; and service quality assurance.

3.4.2 Engagement Approach

Proponents are asked to provide a description of the manner and framework in which the proponent will work with the KRIBHCO to achieve the outcomes. This may include but is not limited to, a description of: assessing and identifying requirements; drafting technical and software architecture; onsite vs. remote work; Agile development or any other software development approach; and any other aspects of the engagement. Proponents are asked to identify the collaboration tools they will utilize for this engagement, bearing in mind the KRIBHCO preferred toolset.

3.4.3 Phases, Sub-Phases, Activities and Timing

Keeping in mind the KRIBHCO's timeframe, Proponents must provide a high-level overview of their proposed phases, sub-phases of execution comprising activities and deliverables, including sequencing and dependencies and estimated approximate timing of each phase, sub-phase or activity.

3.4.4 KRIBHCO's Roles and Responsibilities

Proponents must outline expectations regarding the KRIBHCO's resources required for the proposed engagement approach, the activities each resource will participate in and the estimated time commitment.

3.4.5 Proposed Resources

Proponents must provide details on the proposed team structure and the roles the Proponent will fulfill. The details must include resource's name, their professional title, and the role they will play for the purposes of this specific engagement and whether they are employees of the Proponent or provided through a subcontracting arrangement.

3.4.6 Proposed Resource Bios

Proponents must attach a brief bio of all of the proposed consulting resources, outlining relevant education and experience.

3.5 Financials

Proponents are asked to provide pricing for the overall execution of the project. Proponents may propose a fixed-price for the overall execution of the project.

3.5.1 Pricing

Proponents must provide a firm quote for the overall execution of the project based on the information in this RFP regarding scope and timeline.

Fixed Cost Pricing

Proponents may present one overall fixed price for the project or a modular approach composed of multiple fixed-price elements. Using the table format below, please describe the pricing element, indicate whether or not it is optional, and the fixed price in USD or OMR.

Item	Base/ Optional	Price in INR
Design, Development and Deployment of web-based KHMS (KRIBHCO Hospital Management System) System as per business & technical scope mentioned in Section 2 inclusive of 3 months post go-live support.		
Design, Development and Deployment of web-based KReS (KRIBHCO e-Seva) System as per System as per business & technical scope in Section 2 inclusive of 3 months post go-live support.		
Design, Development and Deployment of Android & iOS based Mobile Application for KReS KRIBHCO e-Seva) System as per System as per business & technical scope mentioned in Section 2 inclusive of 3 months post go-live support.		
Total		

3.5.2 Exclusions

Unless otherwise noted, all information provided in section 3.4 of the RFP, Proposed Services, will be assumed to be included and accounted for in the proposed pricing. Proponents must specify any exclusions.

3.5.3 Conditions and Assumptions

Proponents must document any assumptions made in the preparation of their proposed pricing, and any conditions on which successful completion will be based.

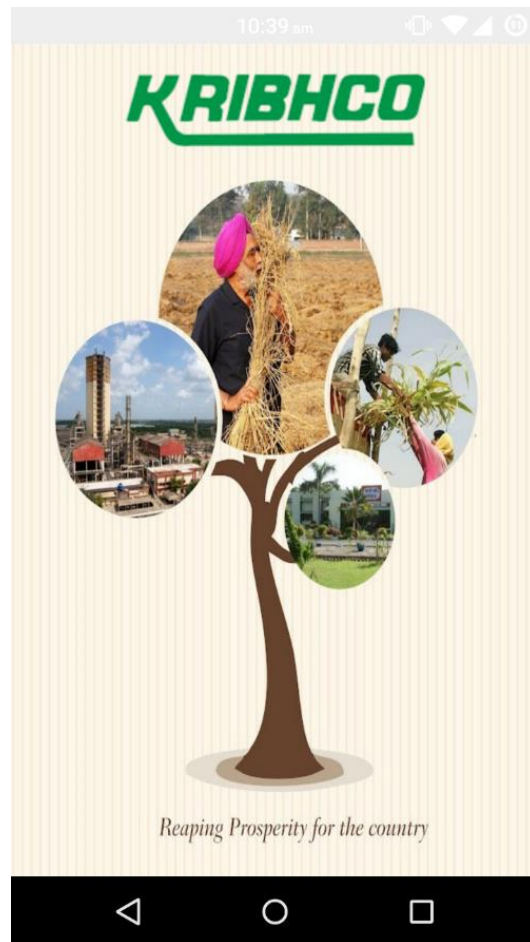
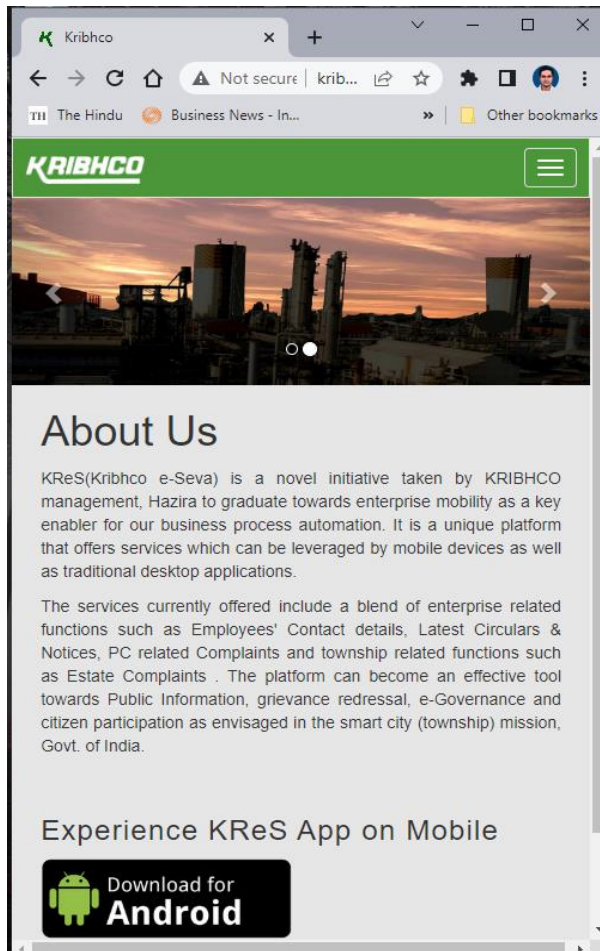
3.5.4 Warranties

Proponents may provide information regarding acceptable options for assurances, guarantees and / or warranties regarding the quality and timeliness of technical services.

Annexure

This section provides an overview of existing KReS Web and Android application as is currently operational in KRIBHCO.

Home Page



Provides an About-Us section along with KRIBHCO plant's wallpaper shot.

Sign In/ Sign Up

The screenshot shows the 'Sign Up' page of the Kribhco website. At the top, there is a green header with the 'Kribhco' logo and a menu icon. Below the header, the page title 'Sign Up' is displayed. The main content area contains a form with a text input field containing '23985' and a 'Go!' button. Below this, a message states 'Sending PIN to registered Mobile Number'. There is a second input field with a blue scribble, followed by 'Go' and 'Cancel' buttons. At the bottom, a green footer contains the text '© Copyright 2016 Krishak Bharati Cooperative Limited'.

The screenshot shows the 'Sign In' page of the Kribhco website. It features a green header with the 'Kribhco' logo and a menu icon. The page title is 'Sign In'. The form includes two input fields: 'Personal Number' and 'PIN'. Below these is a 'Sign in' button. Links for 'Forgot PIN?' and 'Not registered? Sign Up' are provided. The footer is green and contains the text '© Copyright 2016 Krishak Bharati Cooperative Limited'.

The screenshot shows the login screen of the Kribhco mobile app. The background is green with the 'Kribhco' logo at the top. Below the logo are two input fields: 'Personal Number' with a person icon and 'Password' with a lock icon. A large white 'LOGIN' button is centered below the fields. At the bottom, there is a link for 'Forgot Password?'. The Android navigation bar is visible at the very bottom.

Sign Up

The Sign-Up process is used by employees to register themselves on the app. The app uses an employee master table to check for **mobile number** of the employee against their **Personal Number**. It then sends an OTP to verify their identity. Once the employee enters the correct OTP he/she is allowed to create a **PIN** and become a registered user on the application.

Sign In

The user can now use the Personal Number and PIN combination to log on to the application.

Telephone Directory

The screenshot shows a web browser window with the Kribhco logo and a search bar. The search bar contains the text "Name,intercom, qtrno." and a "Go!" button. Below the search bar, there is a table of employee details.

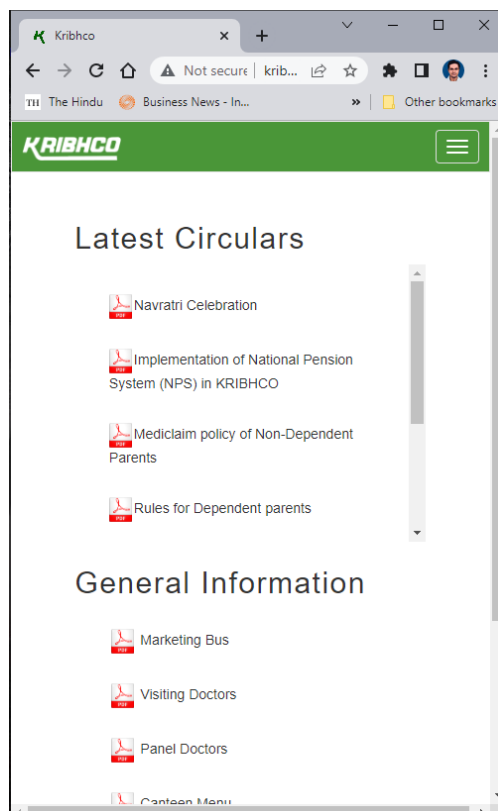
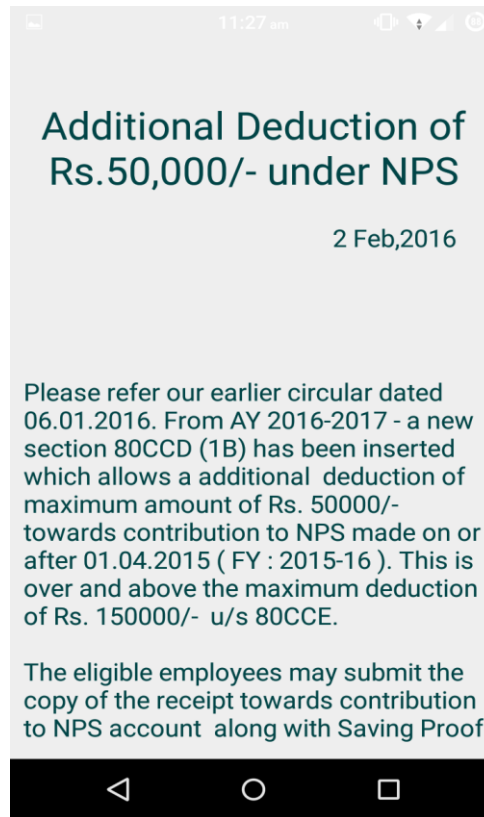
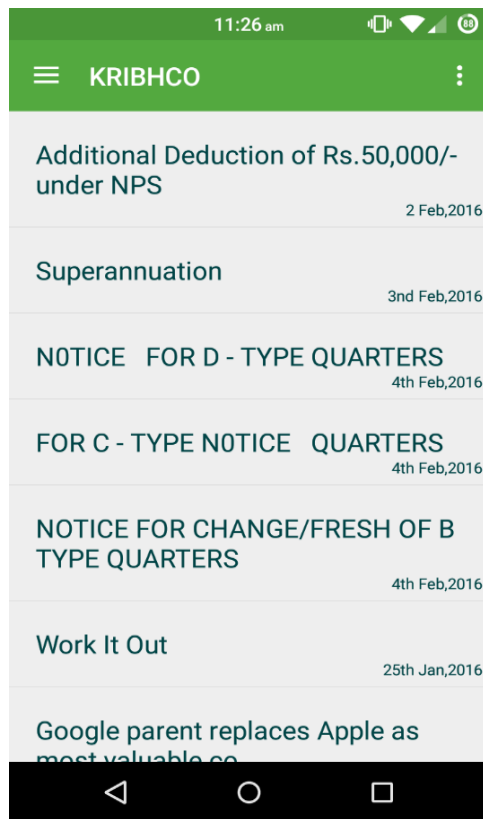
Employee Name
AGARWAL ROHIT
Office Number
Residence Number
QTR Number
Mobile Number
Employee Name
AGARWAL S M
Office Number

The screenshot shows a mobile application interface with a green header bar containing the Kribhco logo. Below the header, there is a list of employee names. A green circular search button is visible on the right side of the list.

Employee Name
Adhikari DN
Agarwal RK
Aggarwal Naman
Aggarwal RK
Ahmed SR
Alok Kumar
Amar Prakash
Ambaliala DD

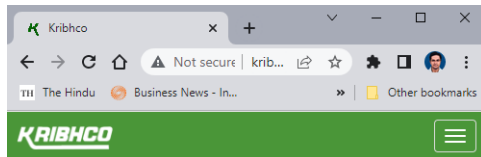
The Telephone Directory displays Employee Contact details which is currently stored in a json file on the server and is accessed using an api. It is proposed that the contact information be stored in database table in the new application.

Circulars



Users can access all the circulars which is maintained by the admin person and application will fetch all the circulars Topic wise and inflate it on the List View of the android application. This circulars service has offline functionality that means user can see these circulars offline also.

PCAMC Complaints

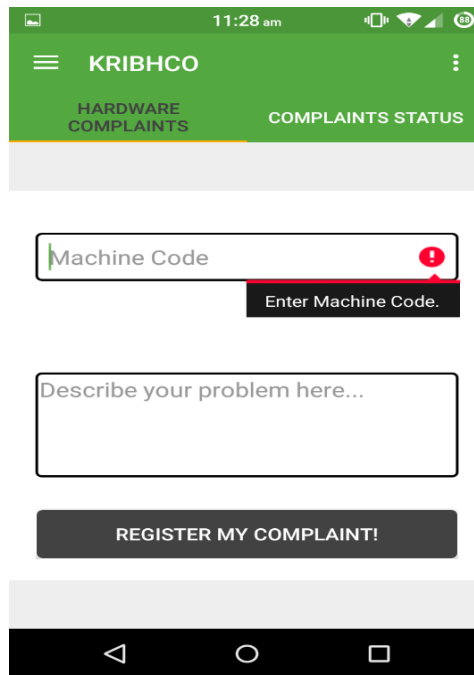
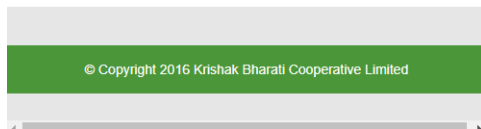


PCAMC Complaints

L1C040

CORE SERIES TFT assigned to ARCHIE KUMAR

Describe your problem here..



PC & Peripherals complaints can be registered using the module.

Estate Complaints

The screenshot shows a web browser window with the Kribhco website. The page title is "Estate Complaints". There are two buttons: "Register Complaint" and "View Complaint Status". Below these, there are radio buttons for "Civil" (selected) and "Electrical". A "Work Type" dropdown menu is set to "Plumbing". There are radio buttons for "Quarter" (selected) and "Bungalows/Other Buildings". A "Quarter Number" input field contains the text "Quarter number". A "Work Requested" section has a text area with the placeholder "Describe your problem here..". A "Submit!" button is at the bottom.

KRIBHCO provides employees quarters to live with their dependents. In township all the different general services regarding household maintenance etc. are managed by the Estate department. Earlier employee or his dependents had to go to the estate office to place complaint manually. However now, employee can place complaint from the KReS application. There are different fields in the application regarding QTR no, location for which place they want to register complaint etc.

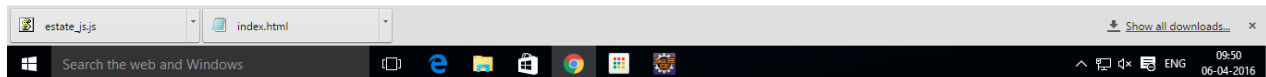
The screenshot shows the Kribhco mobile app interface. The top bar is green with the Kribhco logo. Below it, there are dropdown menus for "CIVIL" and "Carpentry". A list of work types is shown: "Carpentry", "Plumbing", "Mason", "Chocking", "Horticulture", and "Other Jobs". A "Work Requested" section is visible. At the bottom, there is a "REGISTER COMPLAINT" button.

The screenshot shows the Kribhco mobile app interface during the complaint registration process. The top bar is green with the Kribhco logo. Below it, there are dropdown menus for "CIVIL" and "Carpentry". There are radio buttons for "Qtr No." (selected) and "Public Building". A "Qtr.no. eg. 869" input field contains the text "869". A "Bharat Mandir" dropdown menu is visible. A "Location of Building" section is visible. A "Work Requested" section is visible. A green message box says "Your complaint has been registered successfully. Complaint No.C090311". At the bottom, there is a "REGISTER COMPLAINT" button.

Estate Admin



Complaint Number	Complaint Date	Quarter Number	Public Building	Work Requested	Status
E002436	2005-10-06	910	undefined	STREET LIGHT REPLACE	Closed
E003225	2005-11-18	910	undefined	TUBE REPAIR	Closed
E005027	2006-02-25	910	undefined	FAN REPAIR	Closed
E005505	2006-03-25	910	undefined	SWITCH BOARD REPAIR	Closed
E006411	2006-05-10	910	undefined	TUIBE REPLACE	Closed
E006770	2006-05-30	910	undefined	GYSER REPAIR	Closed
E007316	2006-06-26	910	undefined	NO POWER	Closed
E007319	2006-06-26	910	undefined	NO POWER	Closed
E007321	2006-06-26	910	undefined	ELECT WORK	Closed



The Estate Admin interface is used to Update Complaint status and retrieve daily, monthly and annual reports.

PC Admin

ms-2067-amc2 - TightVNC Viewer

[PC Maint. System] - [Complaint Acknowledgement]

File Window Help

Acknowledgement Gate Pass Details...

Complaint No:	U10744
Complaint Type :	
Machine Code:	U1P004
Reported Date & Time:	21/10/2022
Intercom:	Call Priority
Contact Person & Grade:	21280
SUNIL J SAWANT	
PRINTER PAPER CUTTING PROBLEM	
Attended Date & Time:	00/00/0000
Call Report No:	
Category:	
Action taken:	
Engr Name:	
Expected Completion Date :	

Machine Code:	U1P004
Dept Code:	PRODUCTION-UREA-I
Device Code:	04 PRINTER
System Name:	LASERJET PRINTER A4
Model :	HP 1022
Make:	HP
Location:	SHIFT OFFICE
Stand Alone:	
Vendor Code:	SILVER TOUCH TECH.
Inst Dt:	11/10/2006
Exp Dt:	10/10/2007
Amc From:	01/04/2022
Amc To:	31/03/2023
Kribhco Hardware:	☒
Days Under Maint:	365
Stat:	AMC
Machine SLNo.:	
Current Rate:	400.00

Retrieve Records(s)

The PC Admin interface is used to update and extract reports regarding PC Complaints.

KRIBHCO Hospital Management System (KHMS) Overview

